

Subject Description Form

Subject Code	EE502
Subject Title	Modern Protection Methods
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Student should have some prior knowledge in Power Transmission and Distribution
Objectives	1. To introduce the concept of modern power system protection to students. 2. To integrate theory and practical knowledge of power system protection. 3. To explain the design philosophy and working principle of power system protection. 4. To master the analytical techniques. 5. To apply protective relaying in power systems.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Explain the concept and philosophy on power system protection. b. Apply and adapt applications of mathematics, engineering skills in the analysis, comparison, interpretation of various protection schemes in power systems. c. Integrate and justify techniques to be used in the planning and operation of power system protection. d. Solve technical problems for power system protection. e. Present technical results in the form of a technical report.
Subject Synopsis/ Indicative Syllabus	1. Overview of protection system and its development: General considerations. Components of protection. Structure of protective relays. Unit protection and non-unit protection. Trend of protection development. 2. Fault and transient in power systems: Fault transient behaviour in power systems. Computer simulations of the transient behaviour in power systems. 3. Current and voltage transducers: Sources of errors. Requirements of transducers for measurement and protection. Their features and characteristics under steady state and transient conditions. 4. Protection systems for distribution networks: Protection criteria for distribution systems. Features of directional and non-directional protection schemes for distribution systems. 5. Protection systems for transmission networks: Distance protection system and characteristics. Differential line protection. Phase comparison line protection. Use of line carrier and communication for protection systems. 6. Busbar, transformer and generator protection systems: High impedance and low impedance differential protection schemes. Protection schemes for busbar, transformer, and generator. 7. Digital protection relaying technique: Features of digital protection relay. Digital relay architecture. Digital relaying algorithms. Adaptive and intelligent relays. Recent development.

Teaching/Learning Methodology	Lectures and tutorials are the primary means of conveying the basic concepts and theories. Knowledge on system analysis, design and practical applications are given through case studies, in which students are expected to integrate and justify modern techniques to be used in the planning and operation of power system protection with critical and analytical thinking. Mini-projects and experiments are designed to supplement the lecturing materials so that students are encouraged to take extra readings and to look for relevant information.																																																					
	Teaching/Learning Methodology		Outcomes																																																			
		a	b	c	d	e																																																
Lectures		√	√		√																																																	
Tutorials		√	√		√																																																	
Mini-projects and experiments			√	√		√																																																
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><td rowspan="2">Specific assessment methods/tasks</td><td rowspan="2">% weighting</td><td colspan="5">Intended subject learning outcomes to be assessed</td></tr><tr><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td></tr><tr><td>1. Examination</td><td>60%</td><td>√</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>2. Class Tests</td><td>20%</td><td>√</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>3. Homework</td><td>8%</td><td></td><td>√</td><td>√</td><td></td><td>√</td></tr><tr><td>4. Laboratory, Mini-project, and Report</td><td>12%</td><td></td><td>√</td><td>√</td><td></td><td>√</td></tr><tr><td>Total</td><td>100%</td><td colspan="5"></td></tr></table> The examination and tests assess the technical competence of students in power system protection analysis methods and methods of protection design, planning, and operation. Mini-projects, homework, experiments and written reports assess those on analytical skills, problem-solving techniques, and practical considerations of protection design, as well as technical reporting.							Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed					a	b	c	d	e	1. Examination	60%	√	√	√	√		2. Class Tests	20%	√	√	√	√		3. Homework	8%		√	√		√	4. Laboratory, Mini-project, and Report	12%		√	√		√	Total	100%					
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Student Study Effort Expected	Class contact:																																																					
	▪ Lecture/Tutorial			33 Hrs.																																																		
	▪ Laboratory			6 Hrs.																																																		
	Other student study effort:																																																					
	▪ Laboratory preparation/Project/Report			36 Hrs.																																																		
	▪ Homework /Self-study			30 Hrs.																																																		
	Total student study effort			105 Hrs.																																																		
Reading List and References	Reference books: 1. L. Hewitson, M. Brown and R. Balakrishnan, Practical Power System Protection, Newnes, 2005 2. Network Protection and Automation Guide, Alstom Grid, 2011 3. S.H. Horowitz and A.G. Phadke, Power System Relaying, Wiley, 2014 4. J.L. Blackburn and J. Domin, Protective Relaying: Principles and Applications, CRC Press, 2014 5. A.T. Johns and S.K. Salman, Digital Protection for Power Systems, IEE Power Series, 1995 6. Advancements in Microprocessor Based Protection and Communication – IEEE Tutorial Course, Publication No. 97TP120-0, 1997 7. Power System Protection, Vol. 1, 2, & 3, The Electricity Training Association, 1995																																																					